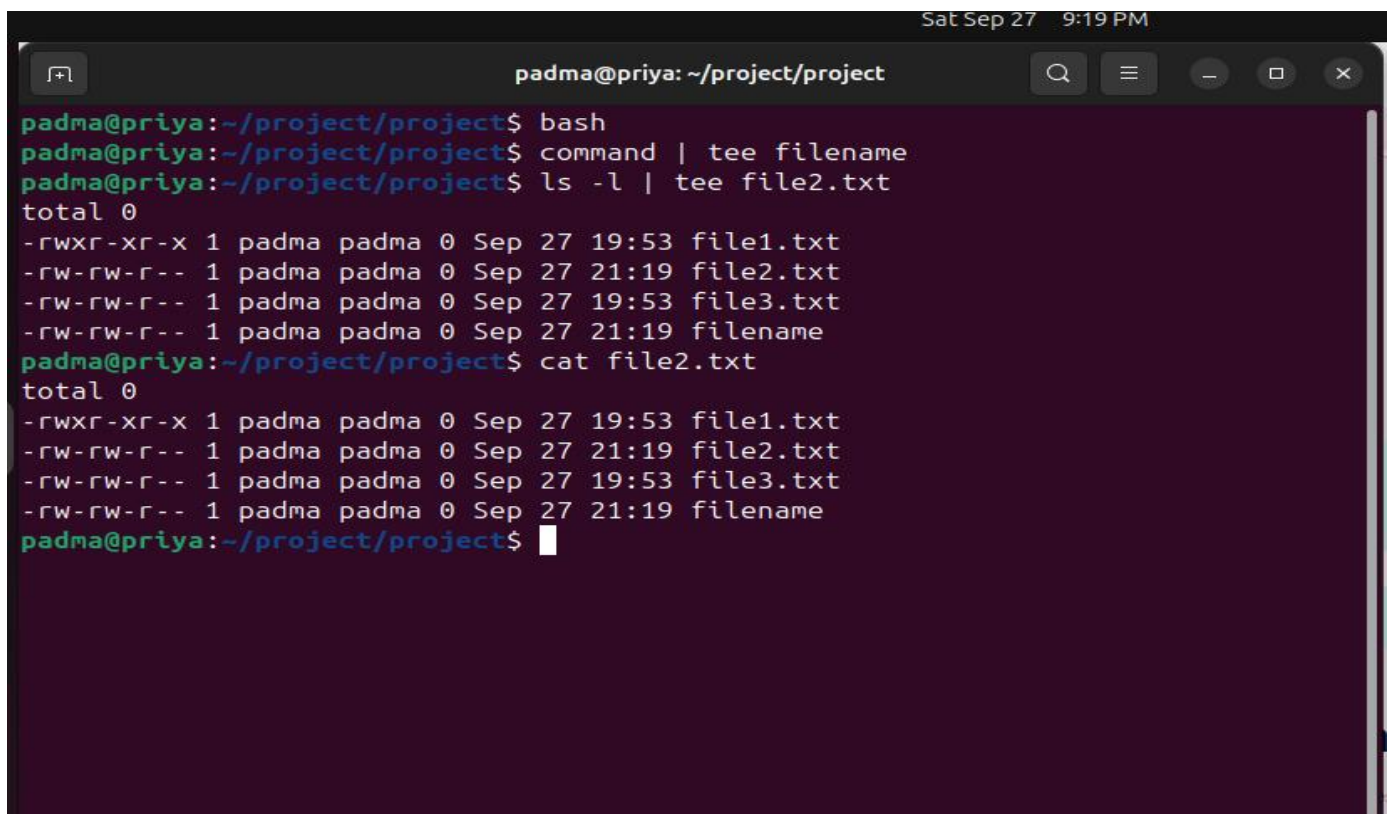


LINUX PROGRAMMING: ASSIGNMENT-3

1. Distinguish between **man** and **whatis** commands? Justify with proper example.

Feature	man	Whatis
Reason	Complete manual page with detail information	Just one line summary
Output	Multi page detail documentation.	Single-line description
Use case	When you want detailed usage, options, examples	When you just want to know what a command does quickly
Command	man command_name	whatis command_name

2. Use the **tee** command to save the output of **ls -l** into a file while also displaying it.



```
Sat Sep 27 9:19 PM
padma@priya: ~/project/project
padma@priya:~/project/project$ bash
padma@priya:~/project/project$ command | tee filename
padma@priya:~/project/project$ ls -l | tee file2.txt
total 0
-rwxr-xr-x 1 padma padma 0 Sep 27 19:53 file1.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 21:19 file2.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 19:53 file3.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 21:19 filename
padma@priya:~/project/project$ cat file2.txt
total 0
-rwxr-xr-x 1 padma padma 0 Sep 27 19:53 file1.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 21:19 file2.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 19:53 file3.txt
-rw-rw-r-- 1 padma padma 0 Sep 27 21:19 filename
padma@priya:~/project/project$
```

3. Explain with an example how the **tee command** can be used in **logging**

- ❖ The tee command in Linux is used to read from standard input and write to both standard output and one or more files simultaneously.
- ❖ This is particularly useful for logging, where you want to see the output on the terminal but also save it to a file for later review.

4. List the steps involved in installing **Ubuntu 25.04 LTS** on Oracle VirtualBox.

❖ Download:

- Virtualbox from (virtualbox.org)
- Ubuntu 25.04 ISO from (ubuntu.com)

❖ Create VM:

- Name: Ubuntu25.04
Type: Linux
Version: Ubuntu (64-bit)
- RAM: 4 GB+
Disk: 25 GB+
VDI
Dynamically allocated

❖ Configure VM:

- Storage → attach Ubuntu ISO
- System → 2+ CPUs, Display → 128 MB video memory

❖ Start VM:

- Boot from ISO → Select Install Ubuntu

❖ Install Ubuntu:

- Language, keyboard, normal installation, erase virtual disk
- Set time zone, user account

❖ Finish & Restart:

- Remove ISO → log in → update system
- Install Guest Additions for better integration

Installation is complete and now you can execute the commands in it.

5. During Ubuntu OS installation, you face a Kernel Panic Error. How would you troubleshoot it?

❖ Steps involved in troubleshoot in kernel panic error in the ubuntu OS:

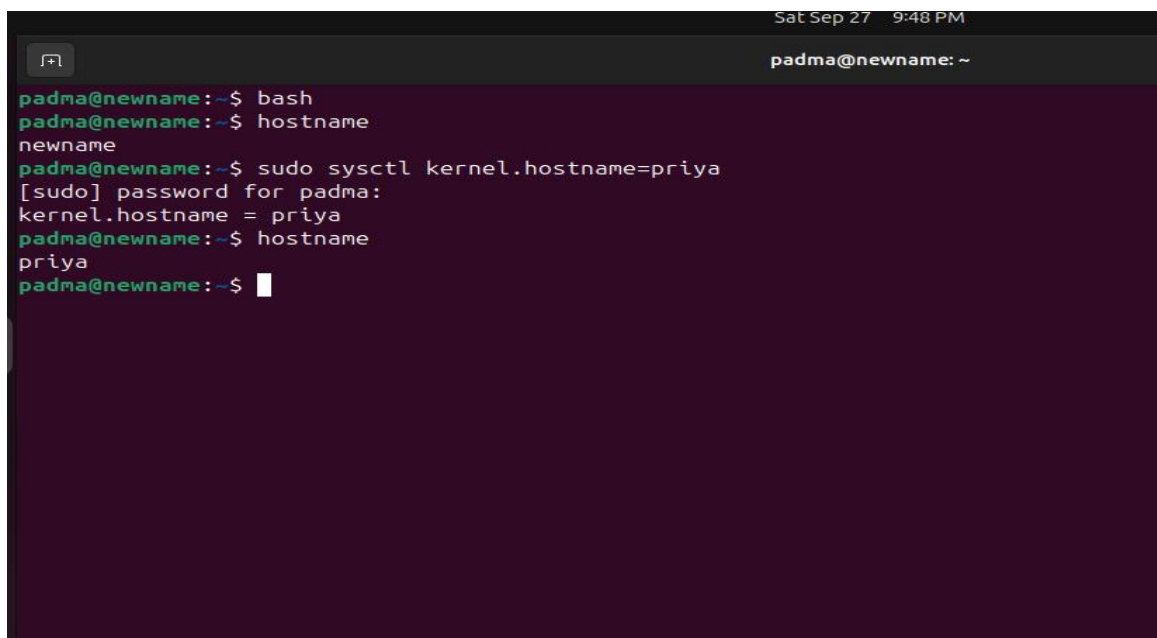
- Check Installation Media:
Verify ISO checksum; recreate USB if needed.
- VM/Hardware Settings:
Ensure enough RAM, CPUs, enable virtualization, correct storage controller.
- Boot Options:
Use nomodeset, acpi=off, or recovery mode.
- Check Hardware:
Test RAM, CPU compatibility.
- Review Logs:
dmesg or /var/log/syslog for errors.
- Reinstall / Try Different Version:
Use stable LTS or alternate ubuntu file.

6. Write the command to display the system's hostname? How to change hostname using sysctl command?

❖ bash

hostname

❖ sysctl kernel.hostname=priya

A terminal window with a dark purple background. The title bar shows 'Sat Sep 27 9:48 PM' and 'padma@newname: ~'. The terminal shows the following commands and output:

```
padma@newname:~$ bash
padma@newname:~$ hostname
newname
padma@newname:~$ sudo sysctl kernel.hostname=priya
[sudo] password for padma:
kernel.hostname = priya
padma@newname:~$ hostname
priya
padma@newname:~$
```

7. Which command is used to show the calendar of the year 1984 with **August** month?

❖ bash

cal command is used to show the calendar

```
Sat Sep 27 9:55 PM
padma@priya: ~
padma@priya:~$ bash
padma@priya:~$ cal 8 1984
    August 1984
Su Mo Tu We Th Fr Sa
                1  2  3  4
 5  6  7  8  9 10 11
12 13 14 15 16 17 18
19 20 21 22 23 24 25
26 27 28 29 30 31

padma@priya:~$
```

8. Write a command to display system uptime and logged-in users together.

❖ bash

uptime;who

```
Sat Sep 27 9:58 PM
padma@priya: ~
padma@priya:~$ bash
padma@priya:~$ uptime
21:58:51 up 2:47, 1 user, load average: 0.39, 1.01, 0.84
padma@priya:~$ who
padma    seat0      2025-09-27 19:43 (login screen)
padma    tty2        2025-09-27 19:43 (tty2)
padma@priya:~$
```

9. Use the find command to list all “.c” files in /home/user.

❖ bash

```
find /home/user -type f -name "*.c"
```

10. How do you change file permissions to allow only the owner to read and write?

❖ To allow only the owner to read and write we use chmod command in linux

❖ bash

```
chmod 600 filename
```

```
chmod u=rw,go= filename
```

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